

Faculty of Engineering and Technology

Department of Electrical and Computer Engineering

Digital Integrated Circuits - ENCS3330

LOW-POWER CMOS D FLIP-FLOP AND 4-BIT REGISTER/COUNTER

Project Report

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Abstract

This project presents the design, simulation, and evaluation of CMOS sequential circuits centered on the D flip-flop as the main storage element. The work begins with reusable CMOS building blocks, including the inverter and two-input NAND gate, which are used to construct a level-sensitive D latch and a positive-edge-triggered baseline D flip-flop. The baseline DFF is then integrated into a 4-bit parallel register and a 4-bit asynchronous binary counter. A separate enable-controlled DFF is also implemented and used to build optimized versions of the register and counter. Electric VLSI is used for hierarchical schematic, icon, and layout development, while LTspice is used to verify functionality and evaluate timing, power, and energy. The project therefore follows a complete baseline-to-optimized custom digital-IC flow and examines how switching activity and clock control influence circuit behavior and energy consumption.

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1. Introduction

Sequential storage elements are fundamental to digital integrated circuits because they preserve state and control how information moves from one clock cycle to the next. The objective of this project is to follow a hierarchical custom-CMOS design flow, beginning with small reusable logic cells, progressing to a D latch and a positive-edge-triggered D flip-flop, and then integrating repeated DFF cells into 4-bit register and counter structures. The second part of the work introduces an enable-controlled DFF and uses it to form optimized register and counter versions.

Electric VLSI was used to create the schematic, icon, and layout hierarchy, while LTspice was used for transistor-level transient simulation and the extraction of timing, power, and energy quantities. The principal simulation supply used in the available tests is VDD = 5 V with the C5 model set. All numerical results reported here are simulation results or layout-related design information, not fabricated-silicon measurements.

1.1 Project Objectives

- Design and verify the CMOS building blocks required for sequential logic.
- Implement a level-sensitive D latch and a positive-edge-triggered baseline DFF.
- Characterize functionality, clock-to-Q delay, average power, and energy, and document the setup/hold measurement method.
- Integrate four DFF cells into a 4-bit parallel register and a 4-bit asynchronous binary counter.
- Implement an enable-controlled DFF and use it in optimized 4-bit register and counter designs.
- Evaluate the influence of switching activity and clock control on measured power and energy.
- Create reusable hierarchical Electric VLSI schematic, icon, and layout cells.

1.2 Design Flow

The implemented hierarchy follows the sequence: basic CMOS gates -> D latch -> baseline DFF -> baseline register/counter -> enable-controlled DFF -> optimized register/counter -> LTspice verification and quantitative comparison. This structure keeps the design modular and makes each stage reusable in the next level of hierarchy.

2. CMOS Building Blocks

2.1 CMOS Inverter

The CMOS inverter is the simplest complementary logic cell used in the project. It contains one PMOS transistor in the pull-up network and one NMOS transistor in the pull-down network. Under stable input conditions, the complementary arrangement provides a full logic-level output with very low static current. In this project the inverter cell is reused for data inversion and clock-phase generation inside larger sequential blocks.

2.2 Two-Input CMOS NAND Gate

The implemented two-input NAND gate uses two PMOS devices in parallel in the pull-up network and two NMOS devices in series in the pull-down network. The NAND cell is a key reusable block in the D-latch implementation and also appears in the control network of the enable-controlled design.

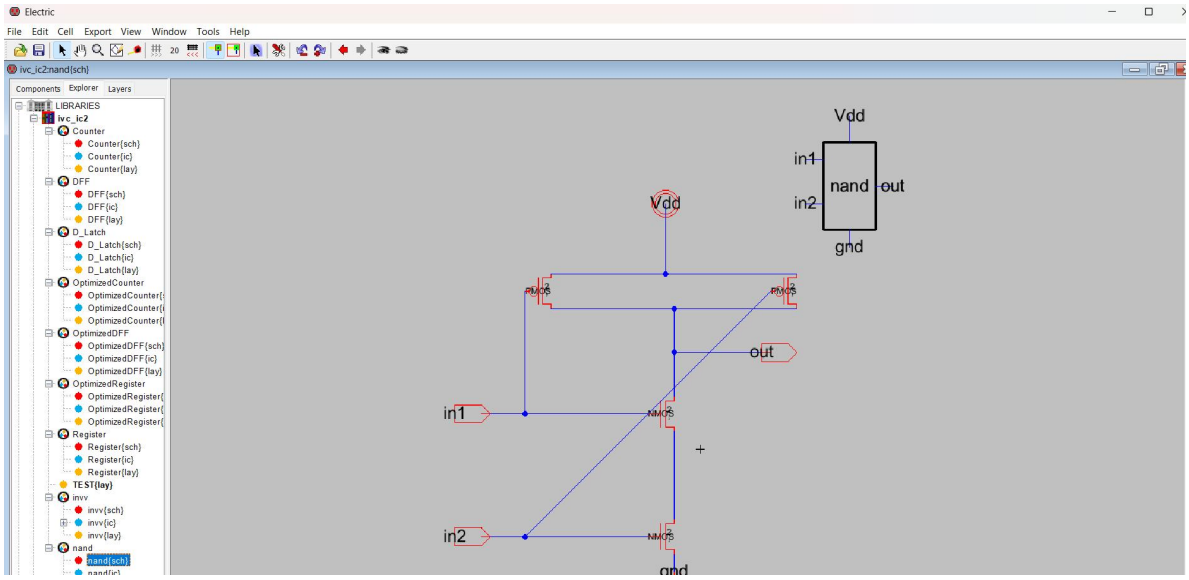


Figure 1. Implemented CMOS two-input NAND schematic in Electric.

3. D-Latch Design

The implemented D latch is an active-high level-sensitive storage element with inputs D and E (enable) and complementary outputs Q and Q'. An inverter generates the complement of D. Two NAND gates form the input control stage, and two cross-coupled NAND gates form the storage pair. When E is active, the latch is transparent and Q follows D. When E is inactive, the cross-coupled NAND pair retains the previously stored state.

From the implemented hierarchy, one latch contains one inverter and four two-input NAND gates. With 2 MOS transistors in the inverter and 4 MOS transistors in each NAND gate, the latch corresponds to 18 MOS transistors.

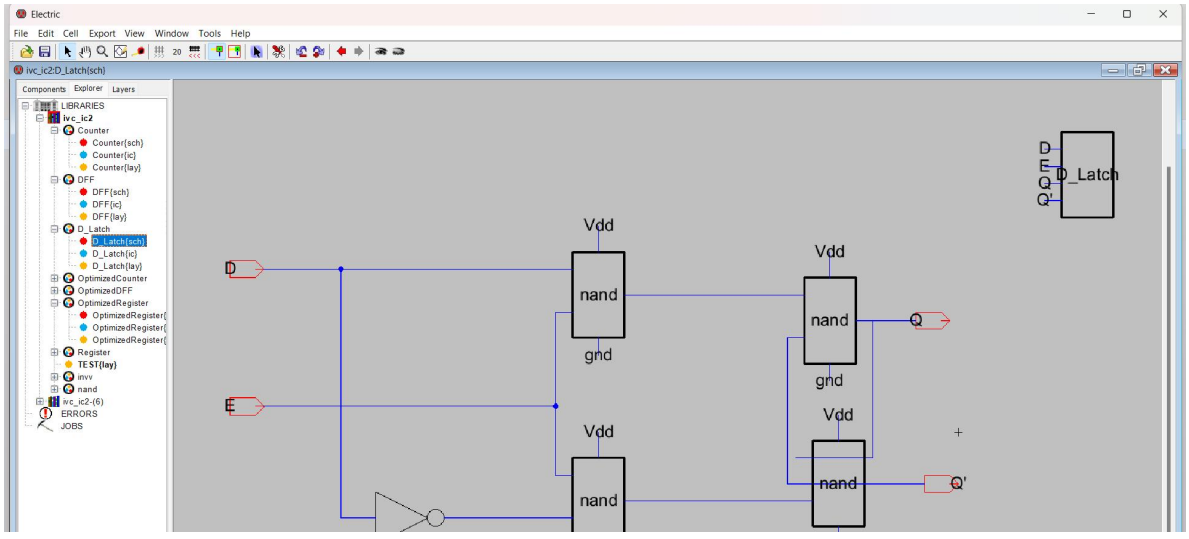


Figure 2. Implemented D-latch schematic showing D, E, Q and Q'.

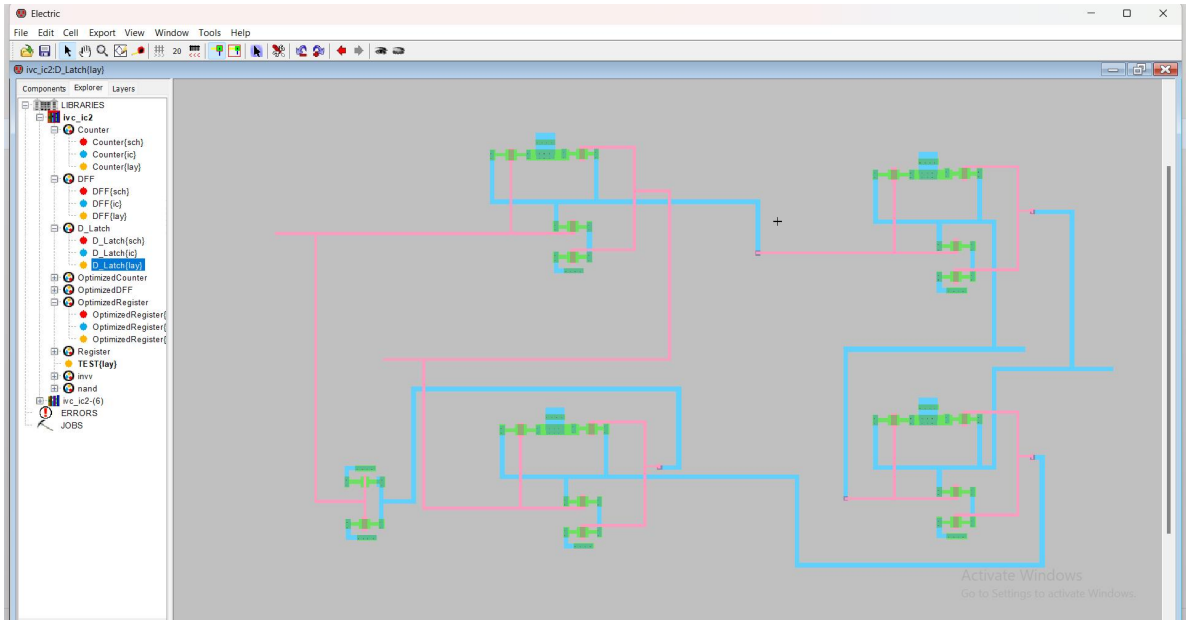


Figure 3. D-latch layout in Electric VLSI.

4. Baseline D Flip-Flop

4.1 Baseline DFF Architecture

The baseline DFF is implemented as a master-slave structure using two D-latch cells and two clock inverters. The first inverter generates the complementary clock phase used by the master latch, while the second inverter restores the clock phase used by the slave latch. This complementary control prevents both latches from being transparent at the same time and produces positive-edge-triggered behavior at the external Q output.

The hierarchical transistor count is 40 MOS transistors: two 18-transistor D latches plus two 2-transistor clock inverters. Q and Q' are both available at the DFF output.

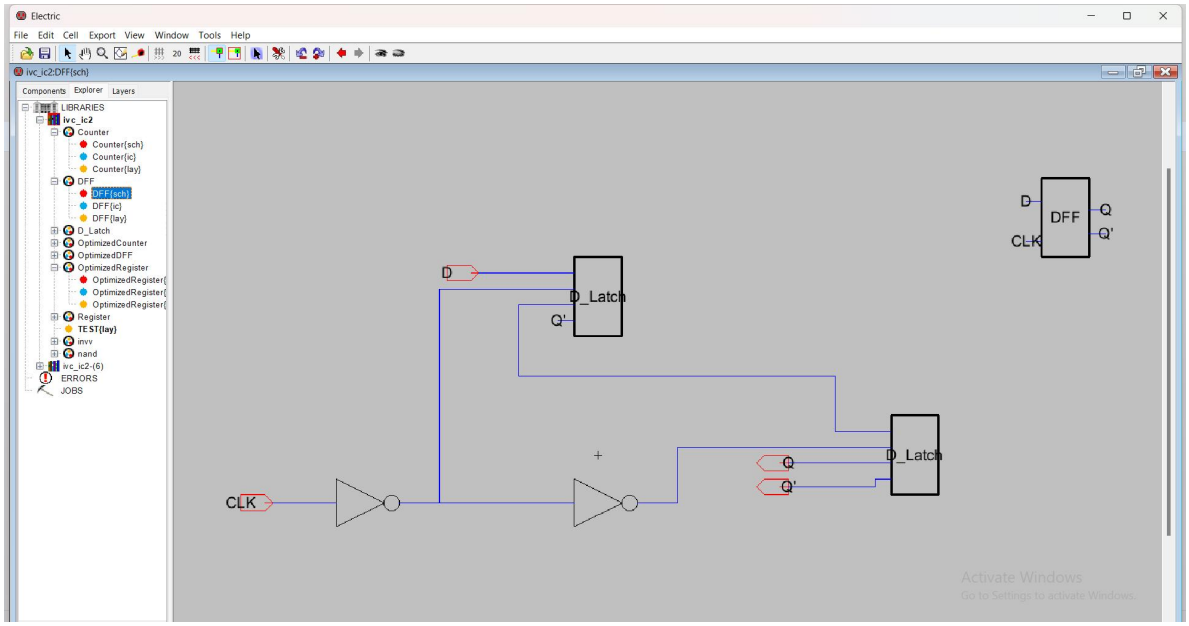


Figure 4. Baseline DFF schematic using two D-latch stages and clock inverters.

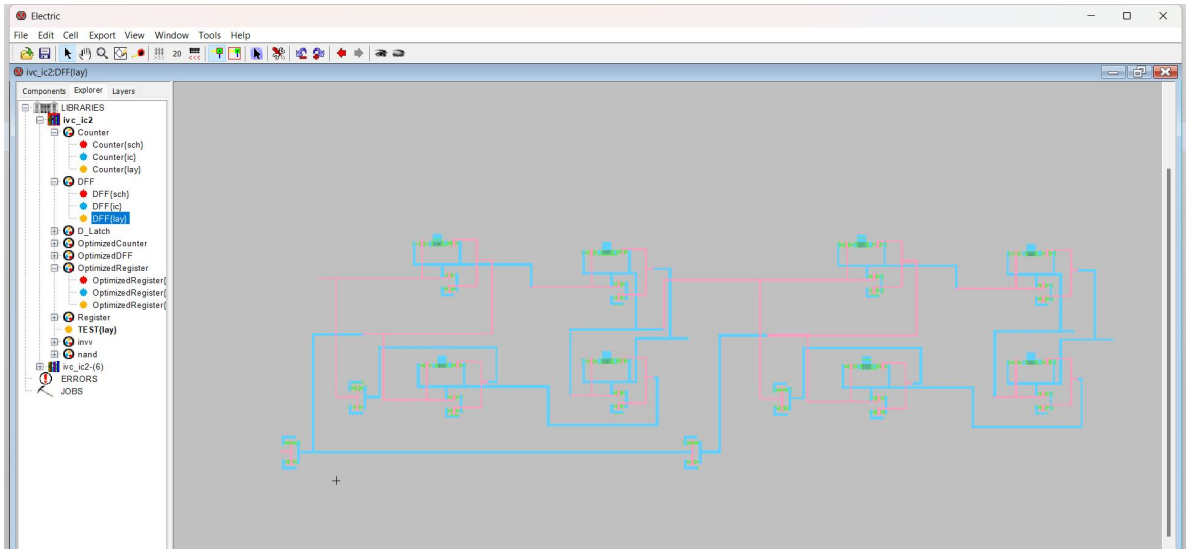


Figure 5. Baseline DFF layout.

4.2 Baseline DFF Operation

During one clock phase the master latch tracks the input while the slave latch holds its previous state. Around the active rising clock edge, the master closes and the slave opens, transferring the stored master value to Q. The external output therefore updates on the positive edge and remains stored between active edges.

5. Baseline 4-bit Sequential Systems

5.1 Four-Bit Parallel Register

The baseline 4-bit register is formed from four identical DFF instances connected to a common clock. Inputs A, B, C, and D are captured in parallel, producing outputs QA, QB, QC, and QD. Because all four storage elements share the same clock, the register performs parallel sampling rather than ripple propagation.

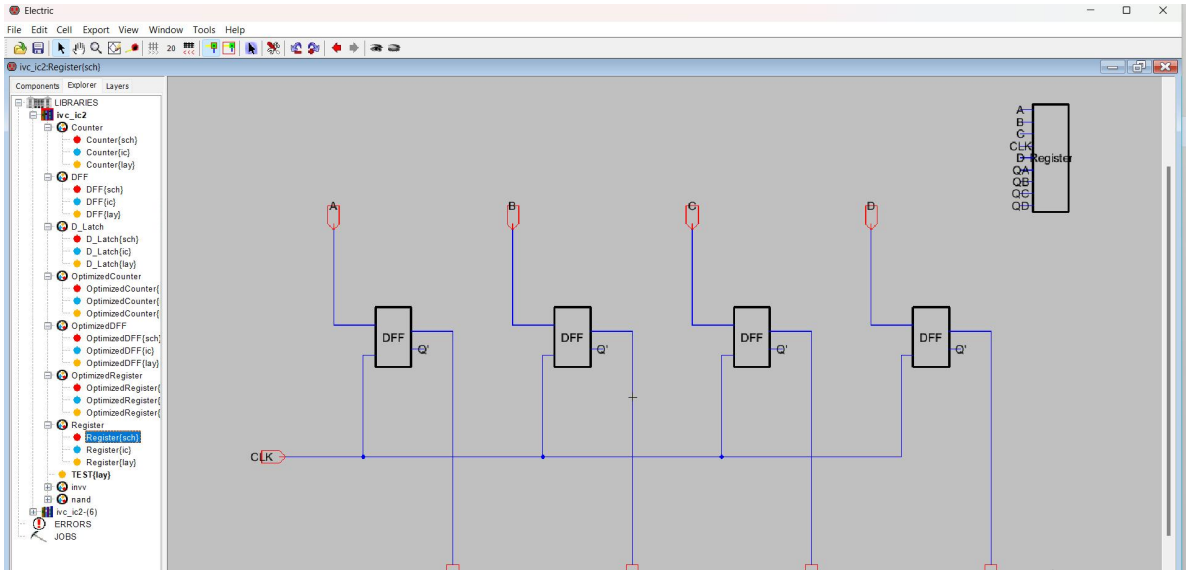


Figure 6. Baseline 4-bit register schematic with four DFFs driven by a common clock.

5.2 Four-Bit Ripple Counter

The baseline counter is an asynchronous 4-bit binary ripple counter built from four positive-edge DFF stages. Each DFF is configured to toggle by feeding Q' back to D. The first stage is clocked by the external clock, while each later stage receives the complementary output of the preceding stage as its clock. Consequently, Q0, Q1, Q2, and Q3 divide the effective toggle rate by 2, 4, 8, and 16, respectively.

The ripple structure is simple and highly reusable, but changes do not occur simultaneously at all bits because the transition must propagate from one stage to the next. The available counter simulation uses initial conditions to start the sequence at 0000; these simulation initial conditions are not a physical reset circuit.

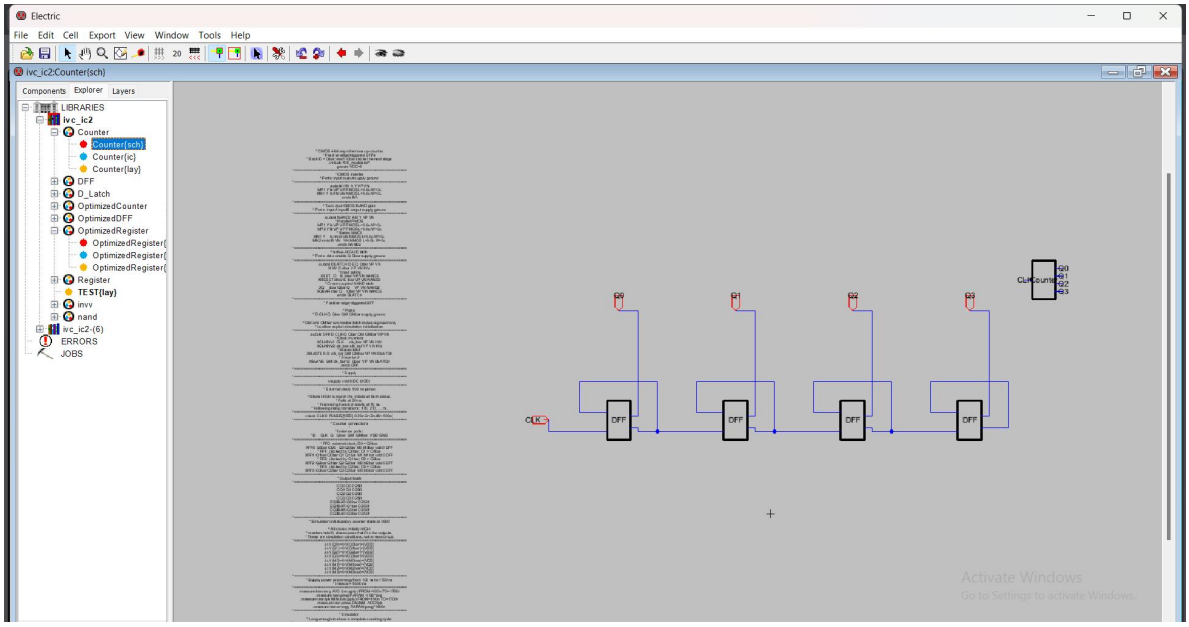


Figure 7. Baseline 4-bit ripple-counter schematic.

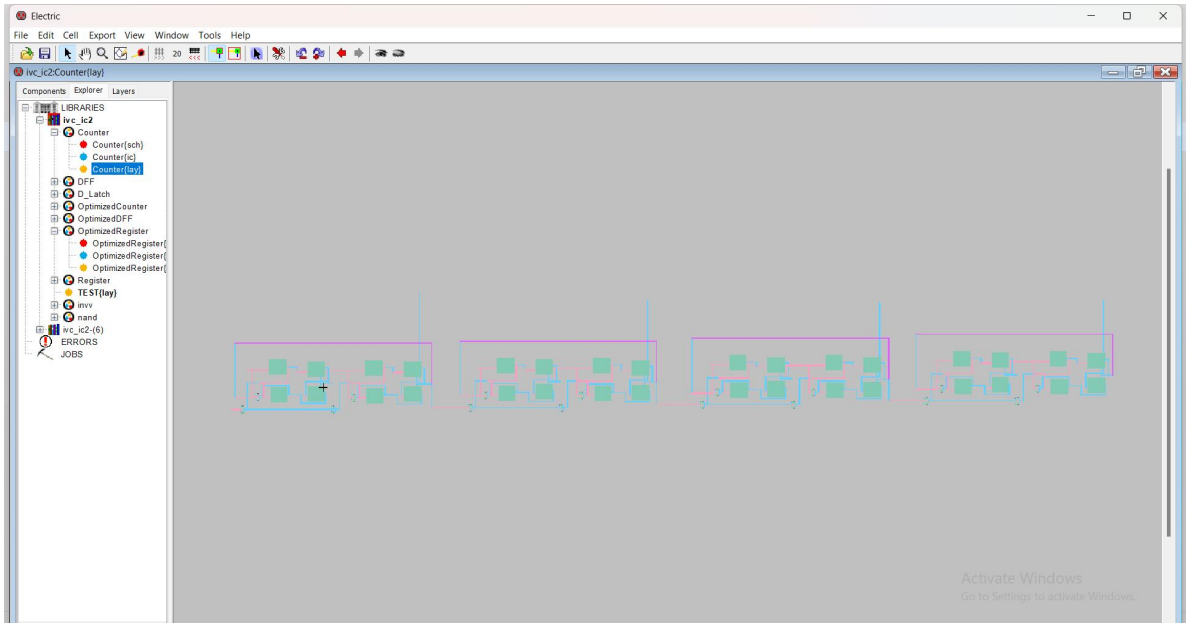


Figure 8. Baseline counter layout using repeated DFF layout cells.

6. Proposed Enable-Controlled Design

6.1 Optimized D Flip-Flop

A separate DFF cell with an enable input EN was implemented as the proposed design. The Electric schematic combines the previously developed latch/DFF hierarchy with additional NAND/inverter control logic in the clock-control path. Functionally, the added EN input allows unnecessary state updates to be suppressed while preserving the stored value when updates are disabled.

The report does not infer a lower-power claim from structure alone. The optimized design is evaluated using its measured waveforms and power results, and any baseline-versus-optimized conclusion is limited to tests performed under comparable conditions.

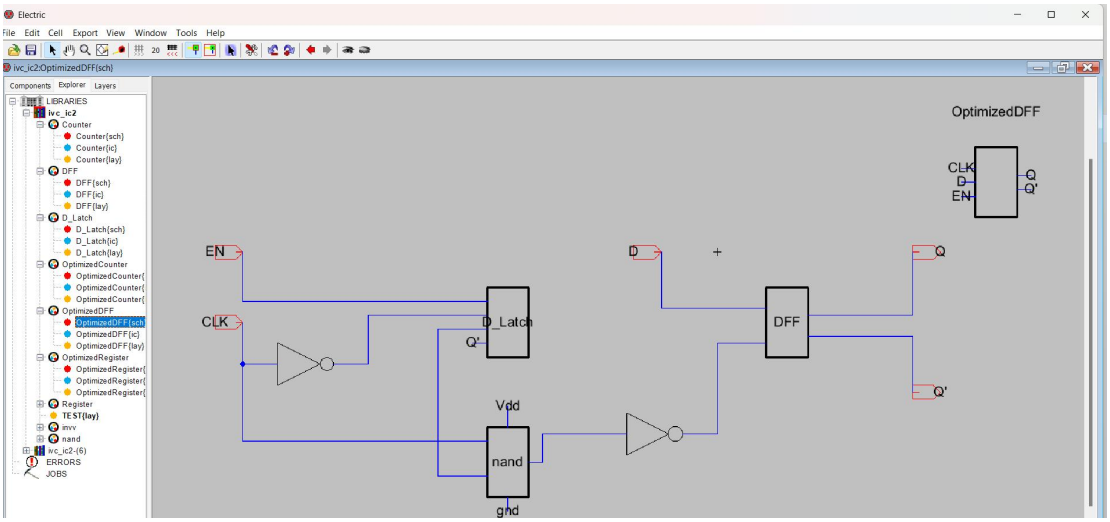


Figure 9. Optimized DFF schematic with D, CLK, EN, Q and Q' interfaces.

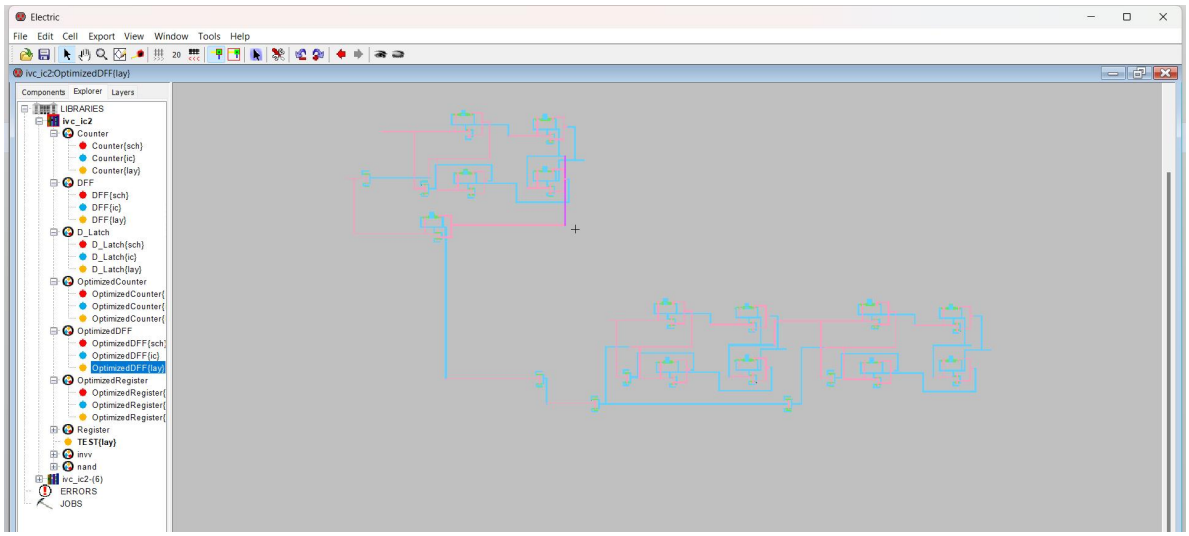


Figure 10. Optimized DFF layout.

6.2 Optimized Four-Bit Register

The optimized register uses four OptimizedDFF instances with common CLK and EN control signals. The four data inputs A-D feed the four storage cells in parallel, while QA-QD provide the stored 4-bit word. The shared enable gives the complete register a common hold/update control.

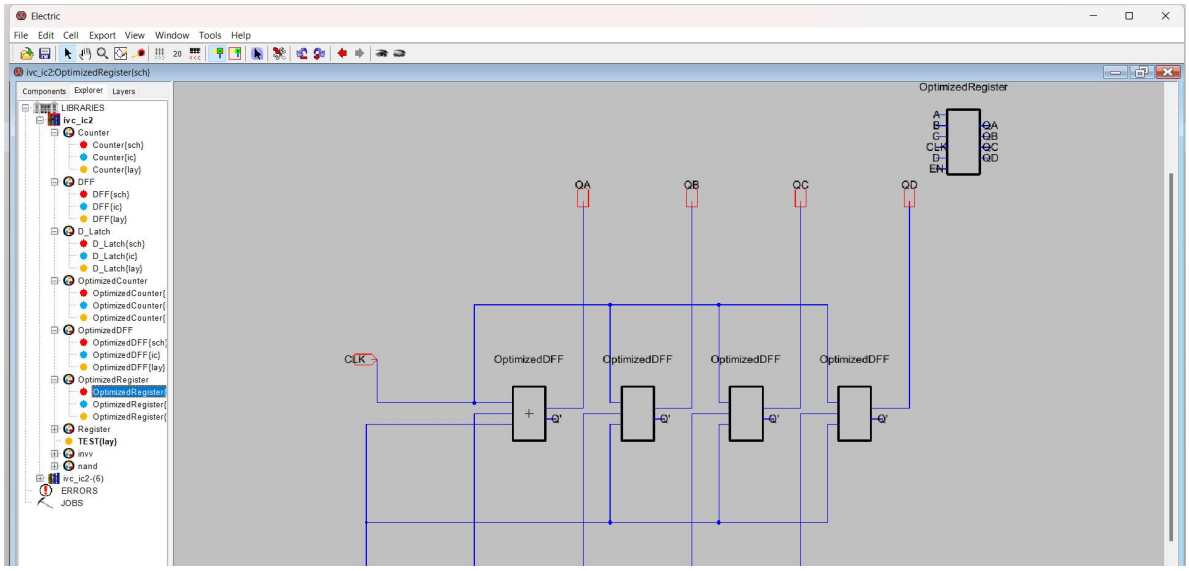


Figure 11. Optimized 4-bit register schematic using four enable-controlled DFF cells.

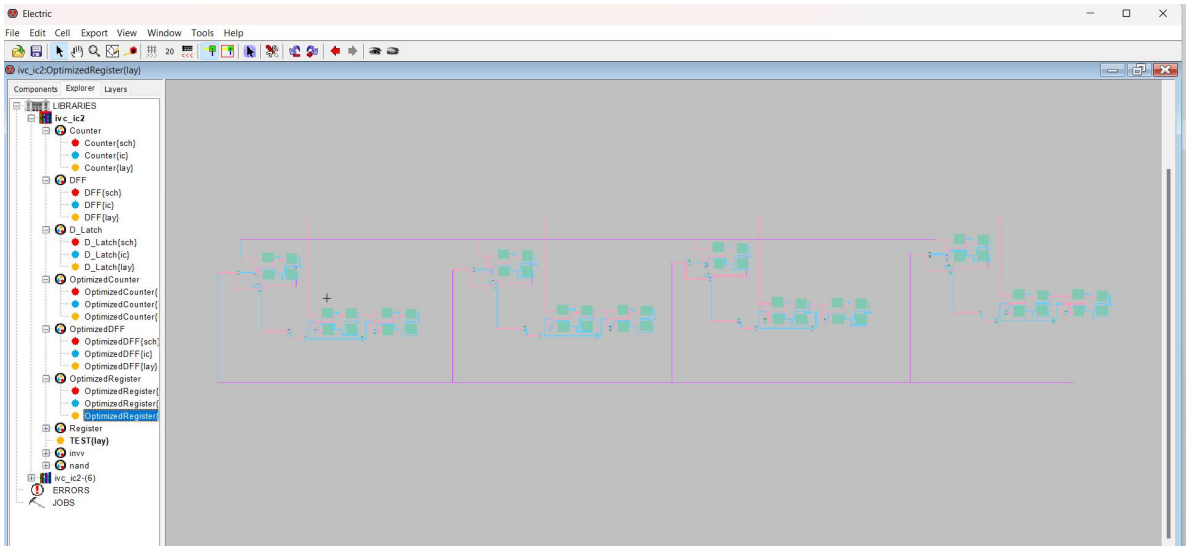


Figure 12. Optimized 4-bit register layout.

6.3 Optimized Four-Bit Counter

The optimized counter uses four OptimizedDFF cells in the same ripple-counting hierarchy, with a common EN control distributed to the stages. The available waveform verifies the intended system-level behavior: the counter runs while EN is asserted, holds its current 4-bit state during the disabled interval, and resumes counting after EN is asserted again.

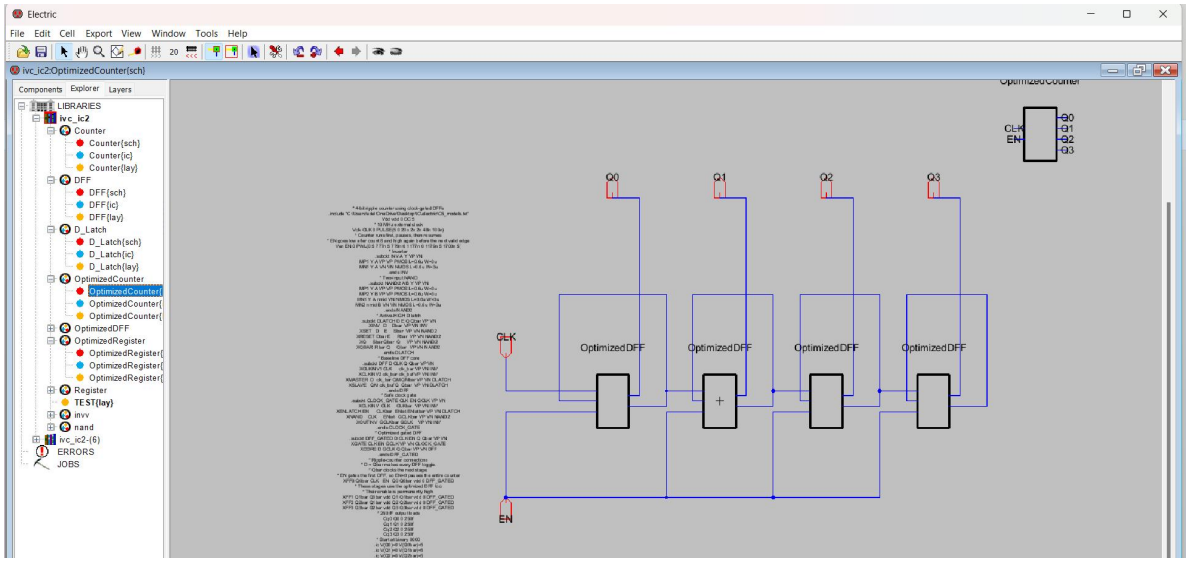


Figure 13. Optimized 4-bit counter schematic with shared EN control.

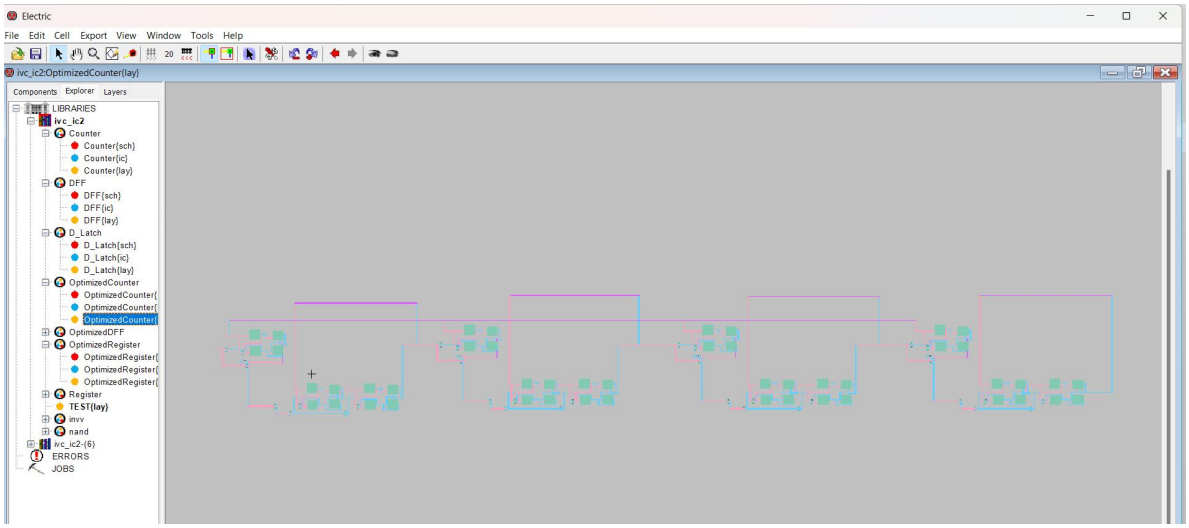


Figure 14. Optimized counter layout.

6.4 Implemented Electric Cell Hierarchy

Table 1. Implemented Electric VLSI cells and available views.

Block	Available Electric Views	Purpose
Inverter (invv)	schematic / icon / layout	Inversion and clock/data buffering
2-input NAND (nand)	schematic / icon / layout	Latch and control logic
D_Latch	schematic / icon / layout	Level-sensitive storage
DFF	schematic / icon / layout	Baseline positive-edge storage cell
Register	schematic / icon / layout	Baseline 4-bit parallel register
Counter	schematic / icon / layout	Baseline 4-bit ripple counter
OptimizedDFF	schematic / icon / layout	Enable-controlled storage cell
OptimizedRegister	schematic / icon / layout	4-bit enable-controlled register
OptimizedCounter	schematic / icon / layout	4-bit enable-controlled ripple counter

7. Simulation and Characterization Methodology

LTspice transient simulation was used to verify functional behavior and to extract the available timing, power, and energy quantities. The supplied simulations use VDD = 5 V and the C5 device-model family. Capacitive loading is included in the counter test, where 250 fF loads are shown on the counter outputs and complementary outputs.

7.1 Functional Verification

DFF verification checks that Q captures D on the active positive clock edge and remains stored between edges, with Q' complementary to Q. Register verification compares the four data inputs with QA-QD at capture events. Counter verification checks the binary count sequence and the expected staged frequency division. The optimized counter additionally verifies run, hold, and resume operation under EN control.

7.2 Timing Measurement

Clock-to-Q delay, tCQ, is measured from the 50% crossing of the active clock edge to the 50% crossing of Q. Separate rising and falling values are retained because the charging and discharging paths are not identical. Setup time was characterized by moving the D transition toward the active rising edge and identifying the capture boundary at the 50% VDD level. Hold time was characterized by moving the D transition through and beyond the active edge. The conservative characterized values obtained are 0.25 ns for setup time and -0.25 ns for hold time.

7.3 Power and Energy Measurement

Average supply current is measured over a stated transient interval. With VDD = 5 V, average power is calculated as Pavg = -5 x Iavg for the LTspice current sign convention used in the testbenches. Energy over the observation interval is then calculated from E = Pavg x Delta t. The same measurement concept is used for the DFF, register, counter, optimized-counter, and NAND tests.

8. Results and Discussion

8.1 Baseline DFF Functionality and Timing

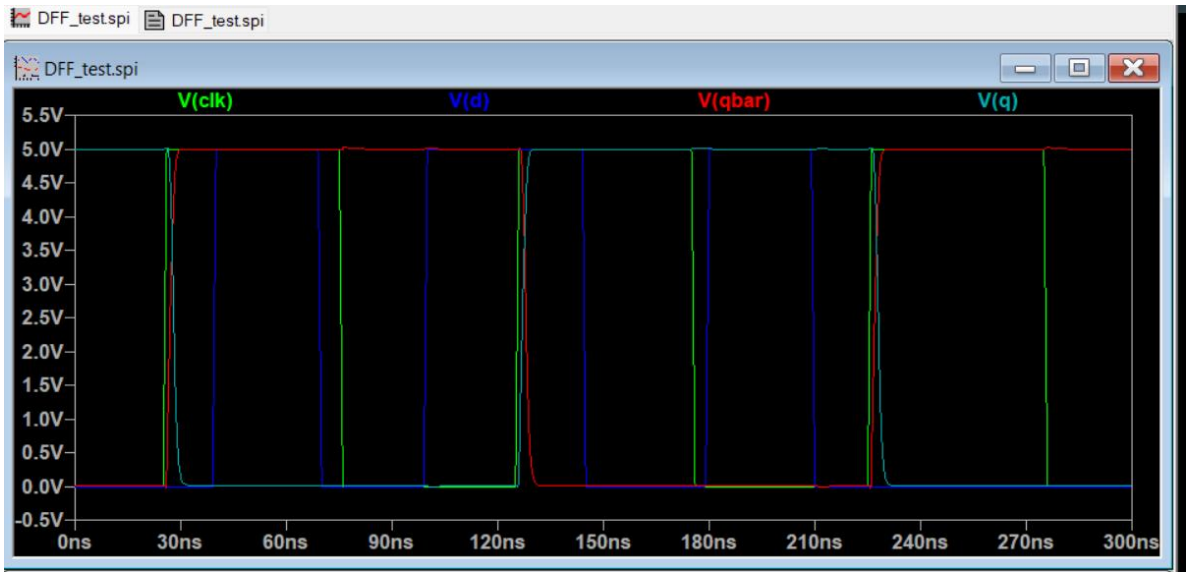


Figure 15. Baseline DFF functional waveform showing CLK, D, Q and Q'.

The waveform confirms the expected positive-edge-triggered behavior. Q and Q' are complementary, Q updates in association with the active clock edge, and the stored output remains stable between active edges.

```
Circuit: * Positive-edge CMOS D flip-flop - ENCS3330 baseline

Vgnd: both pins shorted together -- ignoring.
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tcq_rise=1.45729e-009 FROM 1.255e-007 TO 1.26957e-007
tcq_fall=2.49096e-009 FROM 2.255e-007 TO 2.27991e-007
iavg: AVG(i(vdd))=-2.66644e-005 FROM 1e-008 TO 2.9e-007
pavg: -5*iavg=0.000133322
energy: pavg*280n=3.73302e-011


Date: Sun Sep 13 20:39:16 2026
Total elapsed time: 1.188 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 31123
traniter = 31076
tranpoints = 15173
```

Figure 16. Baseline DFF LTspice timing, average-power and energy measurements.

Table 2. Baseline DFF measured quantities.

Baseline DFF Metric	Measured Value
tCQ, rise	1.45729 ns
tCQ, fall	2.49096 ns
Average tCQ (rise/fall mean)	1.97413 ns
Average supply current magnitude	26.6644 uA
Average power	133.322 uW
Energy over 280 ns	37.3302 pJ

The falling clock-to-Q delay is larger than the rising value in this test. This difference is consistent with unequal effective charging and discharging paths in transistor-level CMOS. The separate rise and fall values are retained as the primary characterization values.

Using the worst-case clock-to-Q delay of 2.49096 ns, the power-delay product (PDP) is approximately 0.3321 pJ.

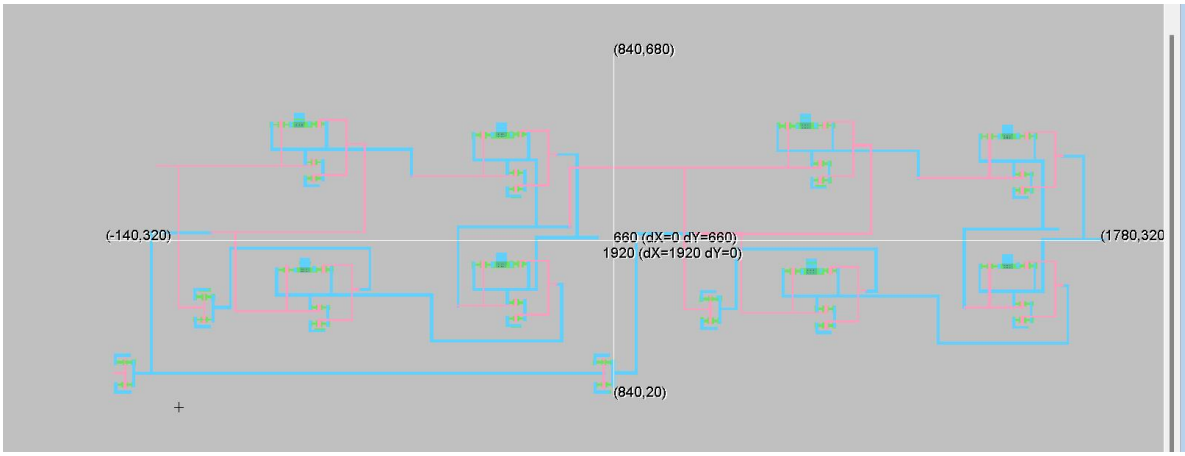


Figure 17. Baseline DFF layout bounding-box measurement in Electric VLSI.

Layout Area Characterization

Baseline DFF layout dimensions = 1920 × 660 Electric layout units

Bounding-box area = 1,267,200 square layout units

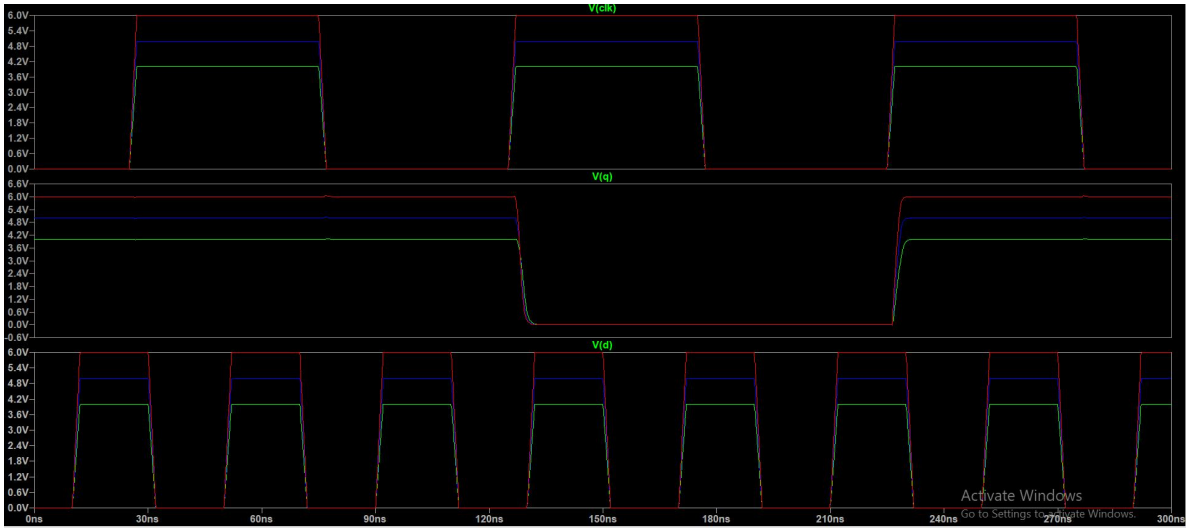


Figure 18. Baseline DFF supply-voltage robustness at 4 V, 5 V, and 6 V.

Supply-Voltage Robustness: The baseline DFF was simulated at supply voltages of 4 V, 5 V, and 6 V. Correct state capture and retention were observed for all three tested supply conditions, confirming functional operation over the selected supply-voltage range.

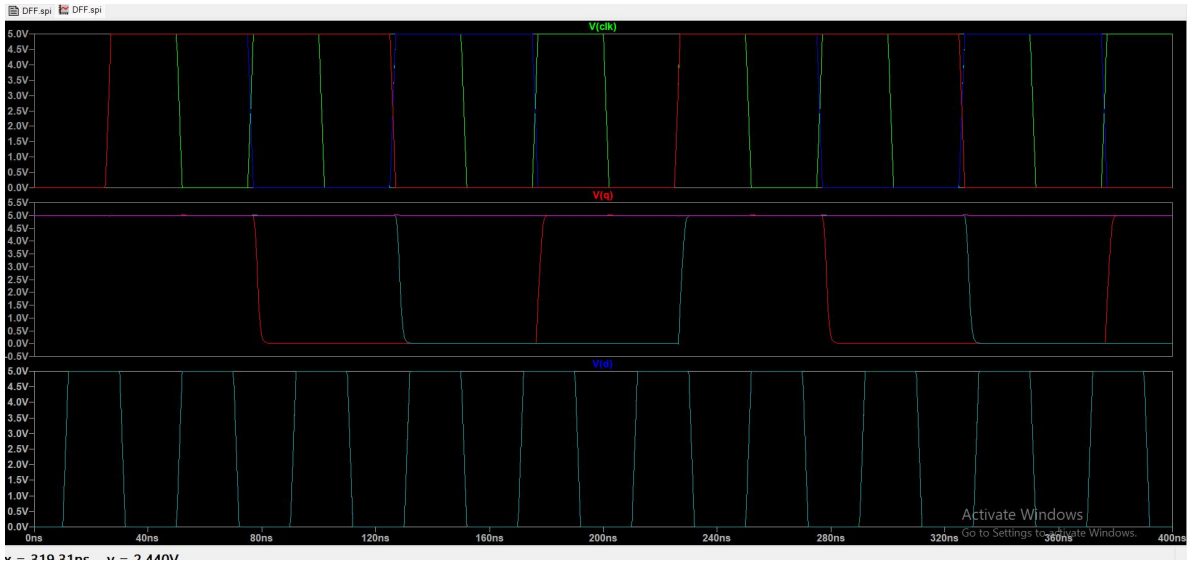


Figure 19. Baseline DFF clock-frequency robustness at 5 MHz, 10 MHz, and 20 MHz.

Clock-Frequency Robustness: The baseline DFF was tested at clock frequencies of 5 MHz, 10 MHz, and 20 MHz while maintaining VDD=5 V and a 250 fF output load. Correct state capture and retention were observed for the selected clock frequencies.

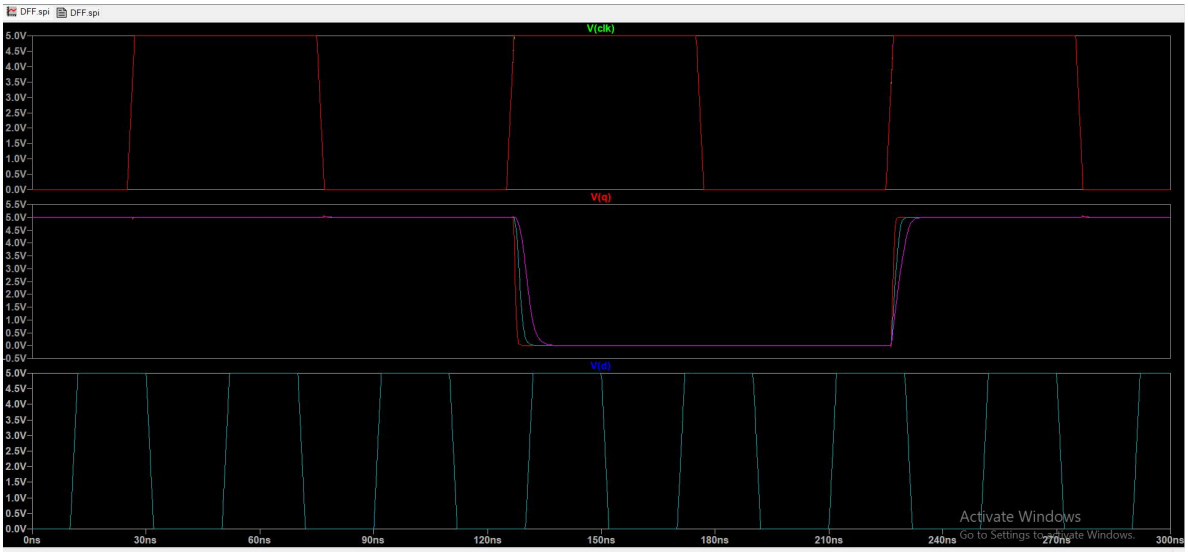


Figure 20. Baseline DFF output-load robustness for 100 fF, 250 fF, and 500 fF.

Output-Load Robustness: The baseline DFF was tested with output loads of 100 fF, 250 fF, and 500 fF at VDD=5 V. Correct state capture and retention were maintained for all tested loads. The simulation also shows that increasing the capacitive load produces a slower output transition, as expected due to the increased charging and discharging time.

Setup-Time Characterization

The setup time was characterized by moving the D-input transition progressively closer to the active rising clock edge. Both the data and clock timing were evaluated at the 50% VDD level (2.5 V). Correct data capture was observed for a 0.25 ns separation, whereas the flip-flop failed when the D and CLK 50% crossings occurred simultaneously. Therefore, the setup-time boundary lies between 0 and 0.25 ns, and 0.25 ns is reported as a conservative setup-time value.

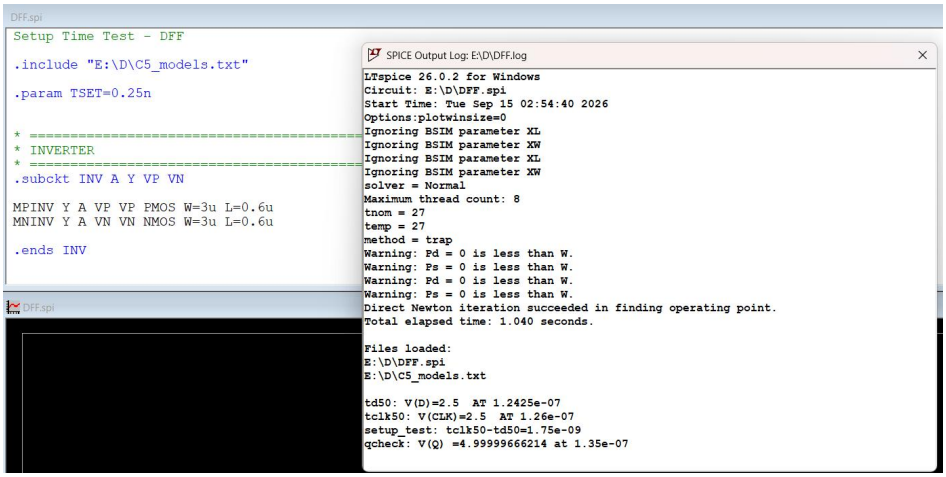


Figure 21. Baseline DFF setup-time characterization showing successful data capture with a 0.25 ns setup interval.

Table 3. Baseline DFF setup-time boundary test.

Setup interval	Result
0.25 ns	PASS
0 ns	FAIL

Reported conservative setup time = 0.25 ns

Hold-Time Characterization

The hold time was characterized by moving the D-input transition progressively toward and beyond the active rising clock edge, using the 50% VDD crossings as timing references. Correct data retention was observed for a hold interval of −0.25 ns, while the flip-flop failed at −0.50 ns. Therefore, the measured hold-time boundary lies between −0.50 ns and −0.25 ns, and −0.25 ns is reported as a conservative characterized hold-time value.

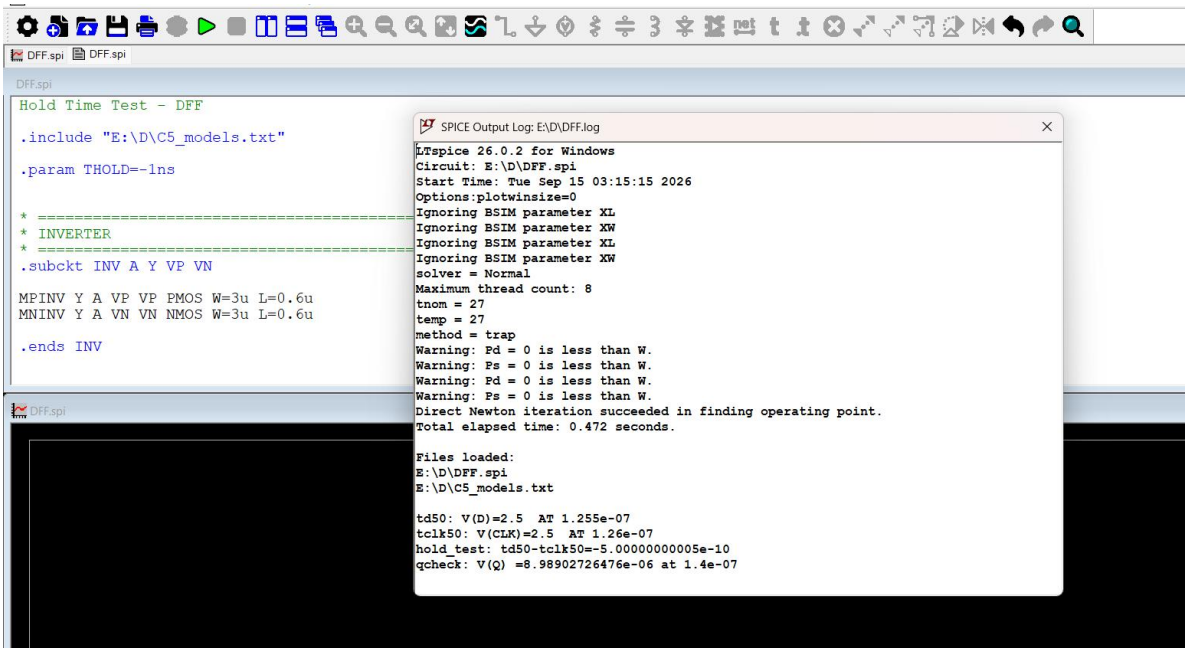


Figure 22. Hold-time characterization of the baseline DFF near the timing boundary.

Setup = 0.25 ns
Hold = −0.25 ns (conservative characterized value)

8.2 Register Results

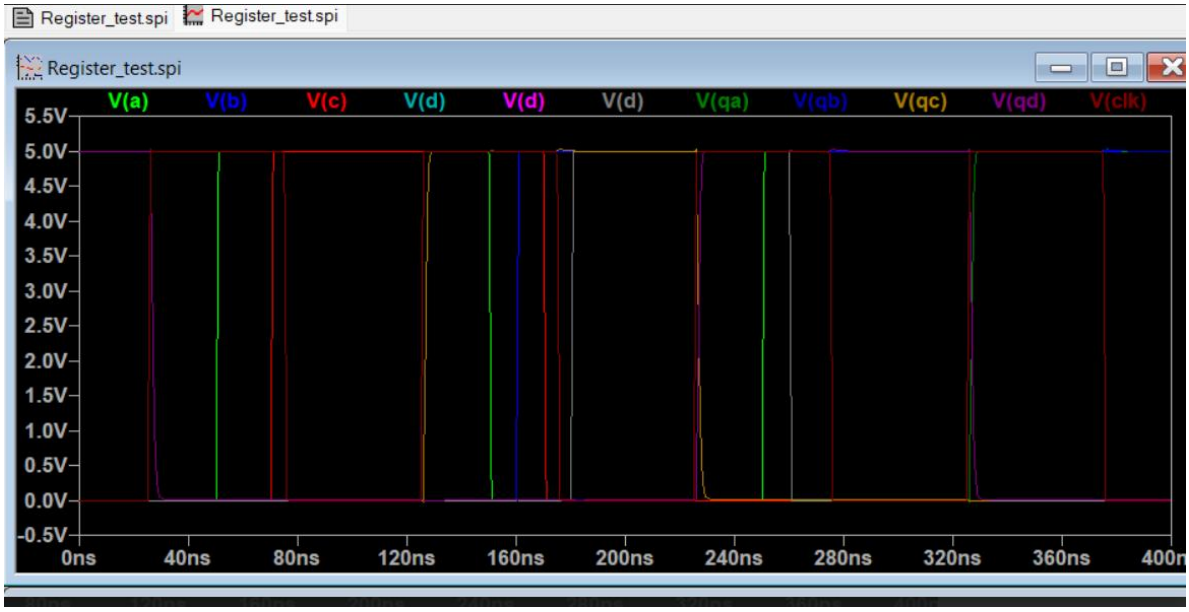


Figure 23. Register input/output functional waveform.

The four register outputs update at clock capture events and remain stored between captures, confirming parallel 4-bit storage.

```
Heightened Def Con from 2.75e-007 ++++++++++++++to 2.75e-007
Heightened Def Con from 3.75e-007 ++++++++++++++to 3.75e-007

iavg: AVG(i(vdd))=-2.49743e-005 FROM 1e-008 TO 3.9e-007
pavg: -5*iavg=0.000124871
energy: pavg*380n=4.74511e-011
```

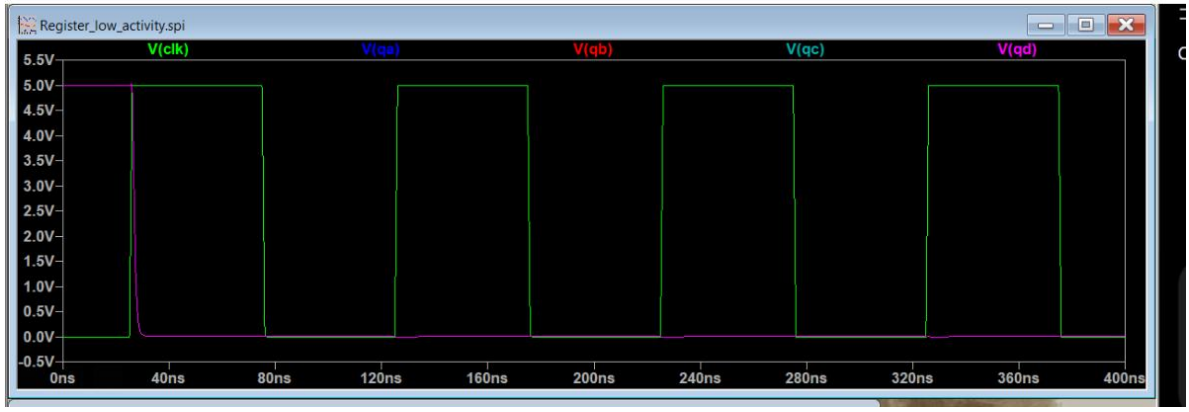


Figure 24. Low-activity register waveform used for activity-dependent power evaluation.

```
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.
Heightened Def Con from 2.75e-007 ++++++++++++++to 2.75e-007

iavg: AVG(i(vdd))=-5.0454e-005 FROM 1e-008 TO 3.9e-007
pavg: -5*iavg=0.00025227
energy: pavg*380n=9.58626e-011


Date: Sun Sep 13 20:49:27 2026
```

Figure 25. Register average-power and energy measurement for the normal activity test.

Table 4. Register activity comparison.

Condition	Pavg	Energy	Observation interval
Normal register test	252.27 uW	95.8626 pJ	380 ns
Low-activity register test	124.871 uW	47.4511 pJ	380 ns

The low-activity register case reduces measured average power by approximately 50.5% relative to the normal register test. Because both tests use the same 380 ns observation interval, the measured energy decreases by essentially the same percentage. This is the clearest supplied result demonstrating the effect of reduced switching activity on dynamic energy consumption.

8.3 Counter Results

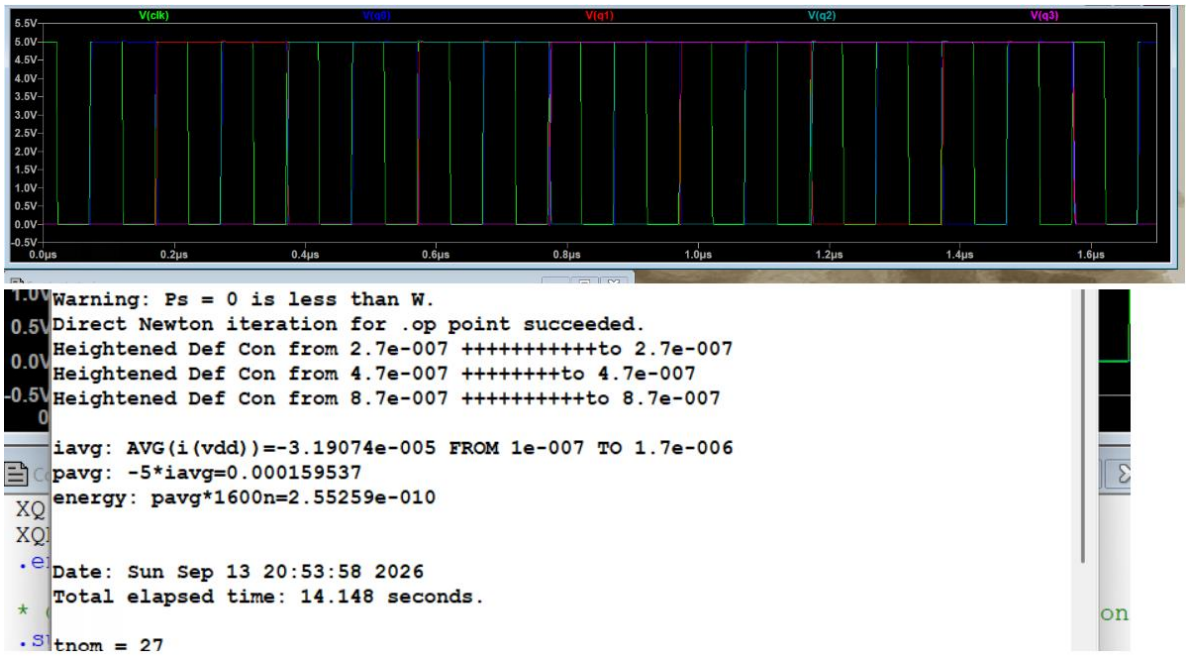


Figure 26. Baseline four-bit ripple-counter waveform and measurement log.

The baseline counter shows the expected staged frequency division, with Q0 toggling most frequently and Q3 least frequently. This verifies asynchronous ripple-counter operation.

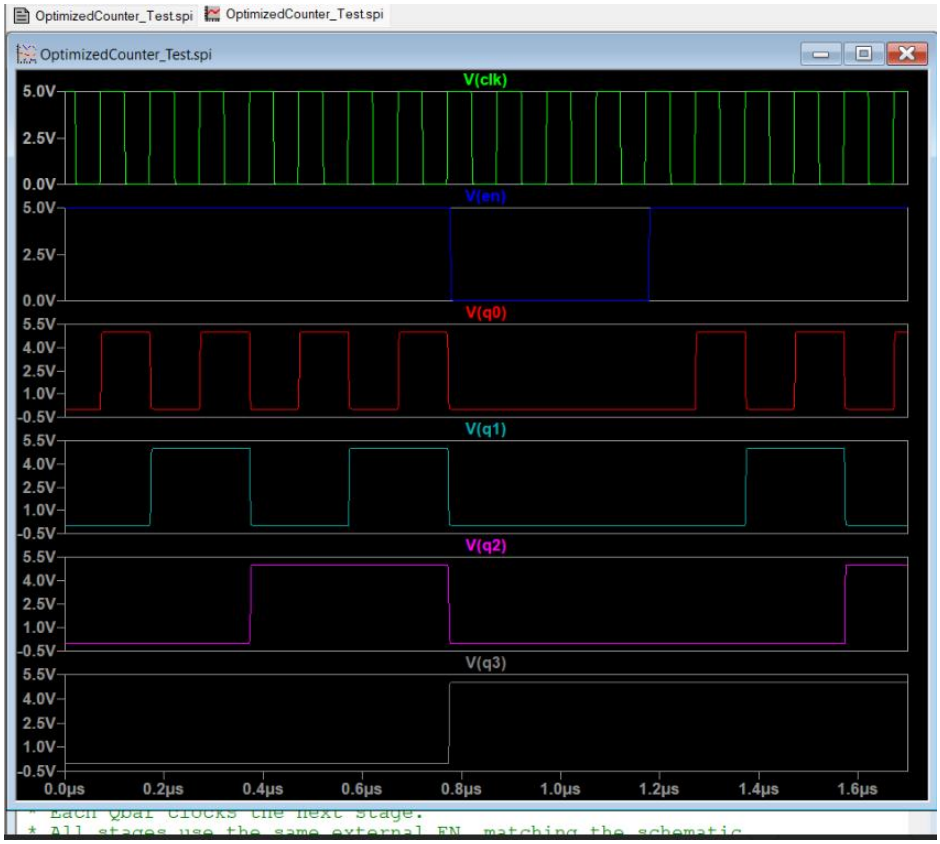


Figure 27. Enable-controlled optimized-counter waveform. The count holds while EN = 0 and resumes when EN = 1.

The optimized-counter waveform verifies the intended enable/hold behavior. In the supplied trace the count reaches 1000 near 0.78 us, remains at that state during the disabled interval, and continues after EN is re-enabled.

```
iavg: AVG(i(vdd))=-3.3284e-005 FROM 1e-007 TO 1.7e-006
pavg: -5*iavg=0.00016642
energy: pavg*1600n=2.66272e-010


Date: Mon Sep 14 02:09:54 2026
Total elapsed time: 22.681 seconds.
```

Figure 28. Optimized-counter average-power and energy measurement.

Table 5. Counter and optimized-counter measurements from the available tests.

Circuit/Test	Average Power	Energy
Baseline counter test	159.537 uW	255.259 pJ over 1.6 us
Enable-controlled optimized counter	166.42 uW	266.272 pJ over 1.6 us

The optimized-counter waveform demonstrates correct EN-based state holding, but the available average-power value for this run is slightly higher than the baseline-counter value. Therefore these two runs do not support a claim of lower average power for the optimized counter. A strict optimization comparison requires identical clock frequency, load, activity sequence, observation interval, and initial conditions.

8.4 NAND Gate Result

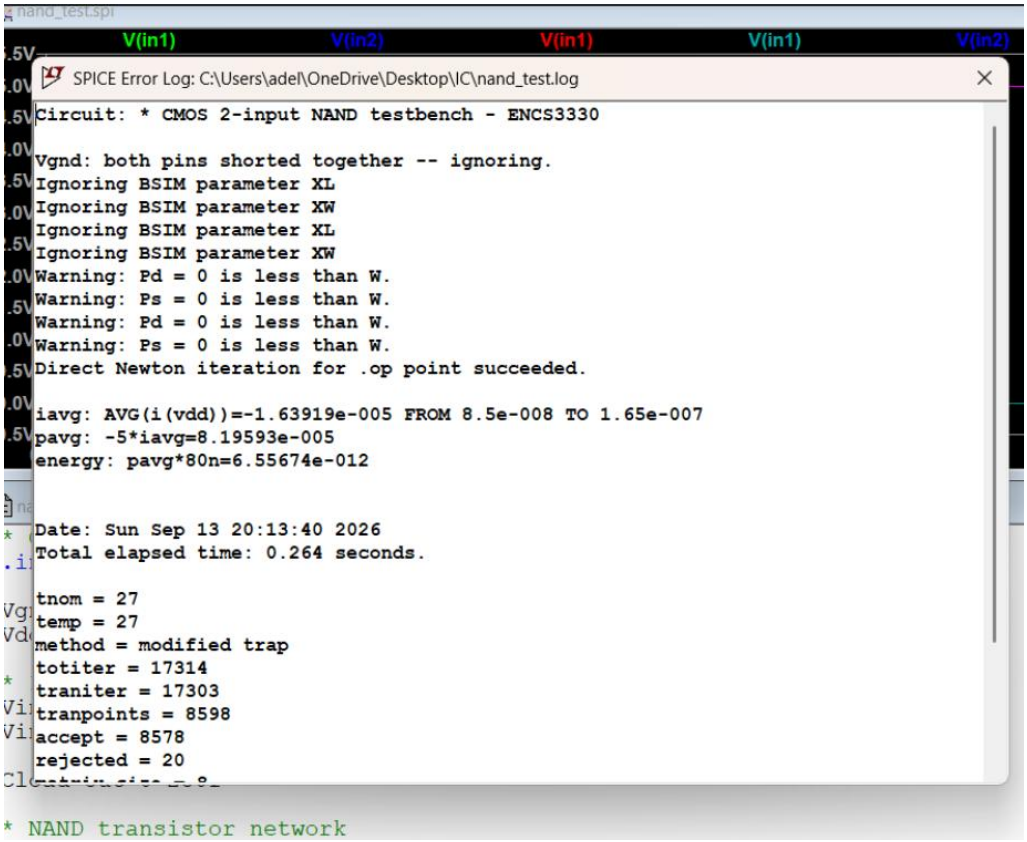


Figure 29. CMOS two-input NAND LTspice average-power and energy measurement.

The NAND-gate test gives an average supply-current magnitude of 16.3919 uA, corresponding to 81.9593 uW at 5 V and 6.5574 pJ over the 80 ns measurement interval. This provides an additional check of the same power-measurement procedure on a basic block used in the D latch.

8.5 Overall Comparison

Baseline and Optimized DFF Comparison

For a fair comparison, the baseline and clock-gated DFFs were evaluated under the same operating conditions: VDD=5V, an output load of 250 fF, and the same clock and data activity. For the clock-gated DFF, the enable signal was held high during this test so that all clock edges were applied to the DFF core. The optimized DFF exhibited clock-to-Q delays of 1.59989 ns and 2.63361 ns for the rising and falling output transitions, respectively. Its average power was 224.696 µW, with an energy consumption of 62.915 pJ over the 280 ns measurement interval. Using the worst-case clock-to-Q delay, the corresponding PDP was approximately 0.5918 pJ.

Table 6. Baseline and clock-gated DFF comparison under continuously enabled operation.

Metric	Baseline DFF	Clock-Gated DFF (EN = 1)
tCQ,riset	1.45729 ns	1.59989 ns
tCQ,fallt	2.49096 ns	2.63361 ns
Worst-case tCQ	2.49096 ns	2.63361 ns
Average Power	133.322 µW	224.696 µW

Metric	Baseline DFF	Clock-Gated DFF (EN = 1)
Energy (280 ns)	37.3302 pJ	62.915 pJ
PDP	0.3321 pJ	0.5918 pJ

With EN continuously asserted, the clock-gated DFF does not provide a power advantage over the baseline DFF. Instead, the additional clock-gating circuitry introduces power and delay overhead. This result is expected for the continuously enabled condition because the clock is not being suppressed. The potential benefit of the proposed design should therefore be evaluated under reduced-activity conditions, where clock gating can prevent unnecessary internal switching.

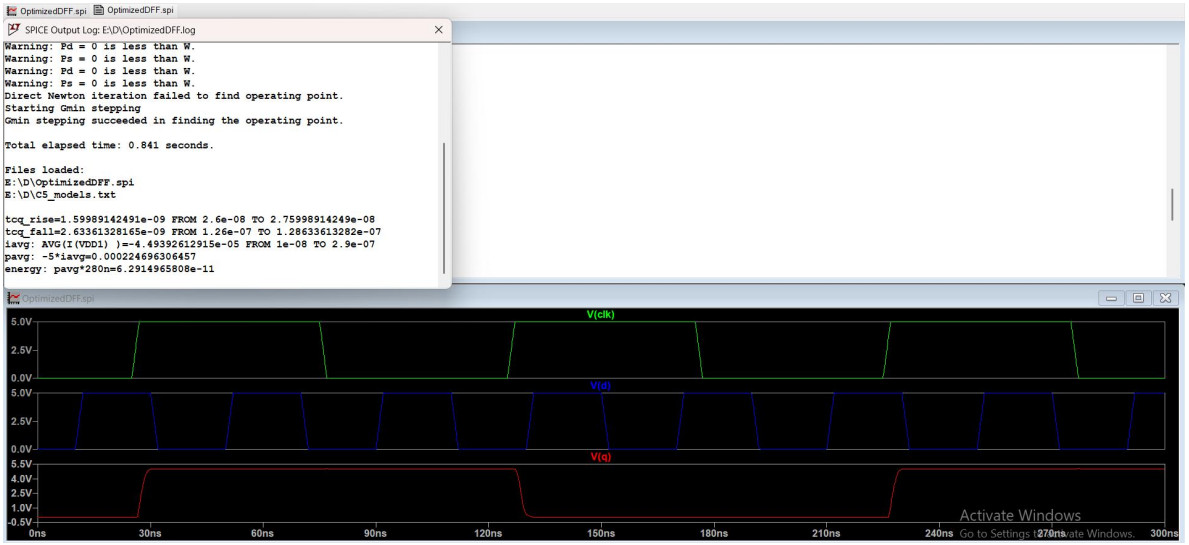


Figure 30. Clock-gated DFF waveform and LTspice measurement under continuously enabled operation.

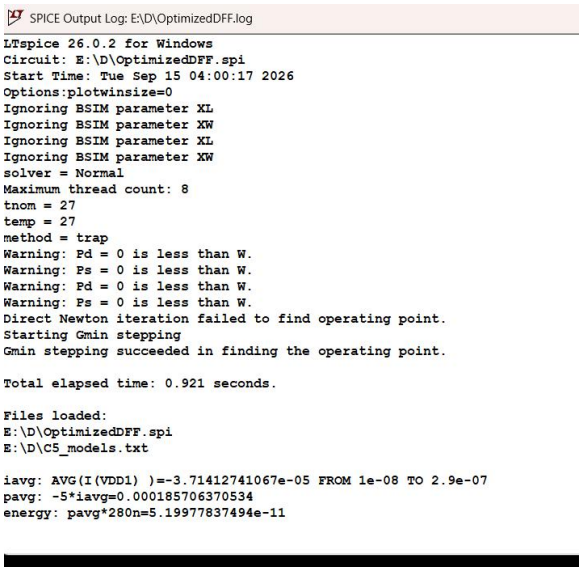


Figure 31. Clock-gated DFF LTspice power and energy measurement under reduced activity.

Clock-Gating under Reduced Activity:

To evaluate the intended benefit of the clock-gating technique, the enable signal was deasserted during a portion of the simulation interval while maintaining the same 5 V supply, clock waveform, data activity, 250 fF output load, and 280 ns power-measurement window. The average power decreased from 224.696 µW in the continuously enabled condition to 185.706 µW under the reduced-activity enable sequence. The corresponding energy decreased from 62.915 pJ to 51.998 pJ. This represents an approximately 17.35% reduction in both average power and energy, demonstrating the benefit of suppressing unnecessary clock activity.

Table 7. Clock-gated DFF reduced-activity power and energy comparison.

Optimized DFF condition	Avg. Power	Energy / 280 ns
EN continuously active	224.696 µW	62.915 pJ
Clock-gated / reduced activity	185.706 µW	51.998 pJ
Reduction	**17.35%**	**17.35%**

Table 8. Summary of available project metrics.

Item	Transistors	Area	tCQ	Setup	Hold	Average Power	Energy
Baseline DFF	40	1,267,200 square layout units	1.457 ns rise / 2.491 ns fall	0.25 ns	-0.25 ns	133.322 uW	37.3302 pJ / 280 ns
4-bit Register	160 (4 x 40-transistor DFFs)	Not measured	Not measured separately	Not measured	Not measured	252.27 uW normal; 124.871 uW low activity	95.8626 pJ / 47.4511 pJ

Item	Transistors	Area	tCQ	Setup	Hold	Average Power	Energy
4-bit Ripple Counter	160 (4 x 40-transistor DFFs)	Not measured	Final-output value not measured	N/A	N/A	159.537 uW	255.259 pJ / 1.6 us
Optimized DFF	66	Not measured	1.600 ns rise / 2.634 ns fall	Not measured	Not measured	224.696 uW (EN=1); 185.706 uW reduced activity	62.915 pJ / 51.998 pJ over 280 ns
Optimized Counter	Not established	Not measured	Not supplied	N/A	N/A	166.42 uW	266.272 pJ / 1.6 us

The comparison table reports only quantities supported by the completed simulations and layout measurements. The baseline DFF now includes measured clock-to-Q delay, setup and hold characterization, power, energy, PDP, and a bounding-box layout area. The optimized DFF includes standalone timing and power/energy results under continuously enabled and reduced-activity conditions. Quantities that were not measured remain explicitly identified as such.

9. Layout Implementation

The Electric implementation uses hierarchical layouts rather than a single flat drawing. The D-latch, baseline DFF, register, counter, optimized DFF, optimized register, and optimized counter are all represented as reusable cells. The repeated-cell organization is especially visible in the 4-bit layouts, where the same storage-cell geometry is tiled and connected by shared clock/control routing.

The report shows representative layout figures alongside the design sections so that each layout can be interpreted next to its schematic. For the baseline DFF, the measured bounding-box dimensions are 1920 x 660 Electric layout units, corresponding to 1,267,200 square layout units. Other layout areas are not reported unless a verified measurement is available. Extracted or post-layout timing is not claimed unless explicitly supported by a corresponding simulation.

10. Limitations

The current result set verifies the main functional blocks and provides baseline-DFF clock-to-Q timing, setup/hold characterization, robustness checks, layout bounding-box area, and several power and energy measurements. Standalone optimized-DFF timing and power were also evaluated under continuously enabled and reduced-activity conditions. Remaining limitations include the absence of verified numerical area for the other layouts, final-output propagation delay for the integrated 4-bit blocks, and extracted post-layout timing. The baseline and optimized counter power values should not be treated as a definitive optimization comparison unless both are repeated with identical activity and test conditions. Finally, the baseline counter uses simulation initial conditions to start from 0000 rather than a physical reset input.

11. Conclusion

This project implemented a complete hierarchical CMOS sequential design flow beginning with inverter and NAND building blocks, followed by a level-sensitive D latch, a positive-edge-triggered master-slave baseline DFF, a 4-bit parallel register, and a 4-bit ripple counter. A separate enable-controlled DFF was then implemented and reused in optimized register and counter structures. Electric VLSI provided the hierarchical schematic and layout environment, while LTspice simulations verified the required sequential behavior and provided timing, power, and energy measurements.

The baseline DFF produced clock-to-Q delays of 1.45729 ns for the rising transition and 2.49096 ns for the falling transition, with an average power of 133.322 uW in the supplied test. The register measurements show the strongest demonstrated activity-dependent saving: reducing switching activity lowered average power from 252.27 uW to 124.871 uW, approximately 50.5%. The optimized counter correctly demonstrated enable-based hold and resume behavior, although its supplied average-power result was not lower than the baseline counter under the reported runs. The results therefore reinforce that low-power conclusions must be supported by controlled comparisons performed under identical operating conditions.

References

[1] ENCS3330 Digital Integrated Circuits, "Low-Power CMOS D Flip-Flop and 4-bit Register/Counter," Project handout, Birzeit University, Summer Semester 2025/2026.

[2] J. M. Rabaey, A. Chandrakasan, and B. Nikolic, Digital Integrated Circuits: A Design Perspective, 2nd ed., Pearson, 2003.

[3] N. H. E. Weste and D. Harris, CMOS VLSI Design: A Circuits and Systems Perspective, 4th ed., Addison-Wesley, 2011.

[4] Electric VLSI Design System, project schematic/layout library and exported SPICE netlists used in this project.

[5] LTspice transient simulation results and measurement logs used in this project.